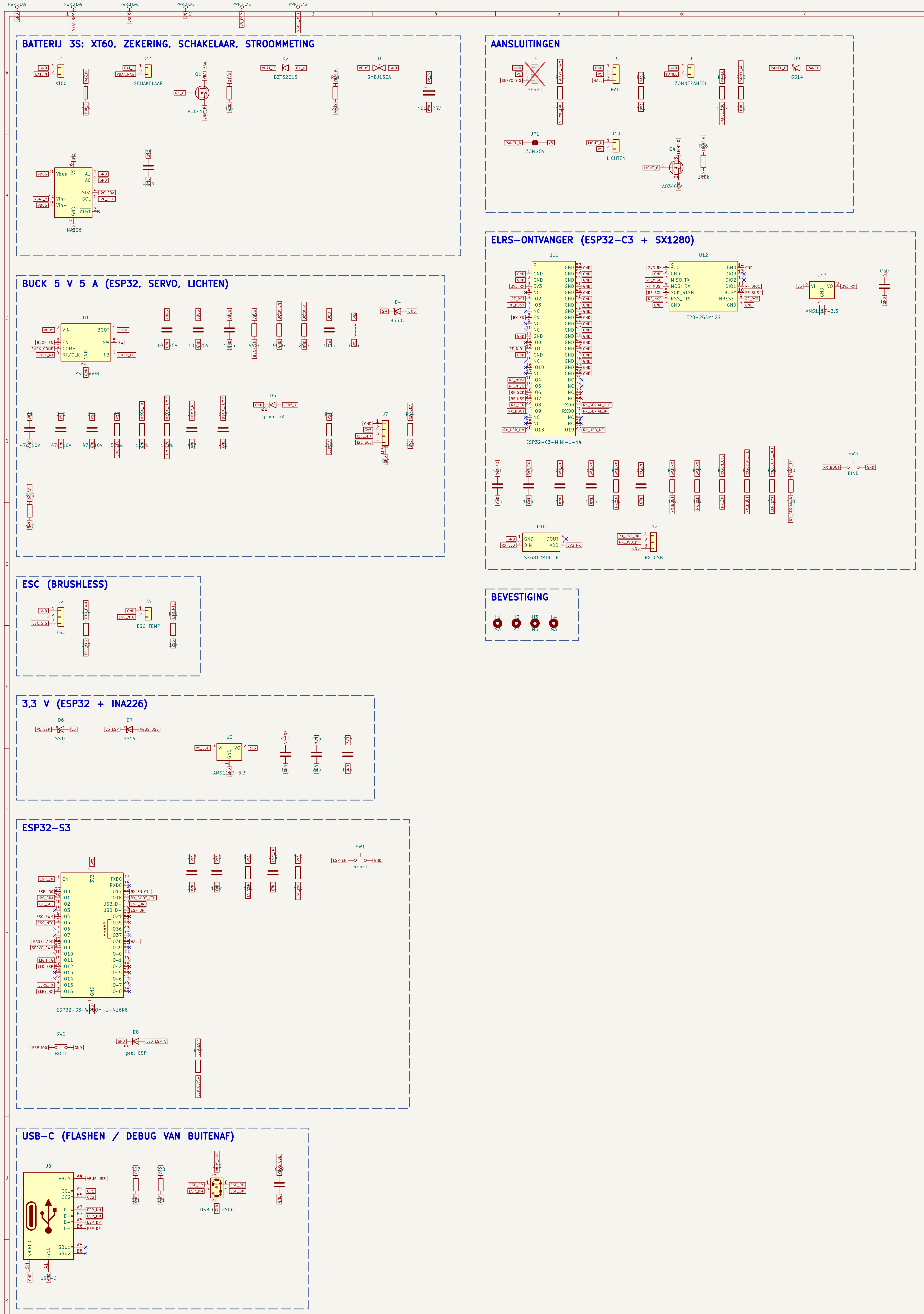
BATTERIJ 3S: XT60, ZEKERING, SCHAKELAAR, STROOMMETING
XT60
SCHAKELAAR
AO4407A
BZT52C15
SMBJ15CA
100u 25V
INA226
AANSLUITINGEN
SERVO
HALL
ZONNEPANEEL
SS14
ZON>5V
LICHTEN
AO3400A
ELRS-ONTVANGER (ESP32-C3 + SX1280)
ESP32-C3-MINI-1-N4
E28-2G4M12S
AMS1117-3.3
BIND
SK6812MINI-E
RX USB
BUCK 5 V 5 A (ESP32, SERVO, LICHTEN)
TPS54560B
B560C
green 5V
ESC (BRUSHLESS)
ESC
ESC TEMP
BEVESTIGING
3,3 V (ESP32 + INA226)
SS14
AMS1117-3.3
ESP32-S3
ESP32-S3-WROOM-1-N16R8
RESET
BOOT
geel ESP
USB-C (FLASHEN / DEBUG VAN BUITENAF)
USB-C
USBLC6-2SC6

SC-02 DRIVE BOARD v3.0 – compact board on M3 standoffs (battery on top): ESP32-S3 + brushless ESC output + INA226 + 5 V 5 A buck + built-in ExpressLRS receiver (generated by tools/build_sch.py from tools/design.py)
Battery: XT60 -> 10 A fuse -> power switch pads -> P-MOSFET reverse protection -> 2 mOhm shunt (INA226 0x40) -> VBUS: motors + buck.
Motor: one brushless 2435 4500KV on a 40 A ESC (battery via Y-cable, not through this board). ESC signal IO4 (100 R), ESC/motor NTC IO5 (10k divider). Servo IO9 (100 R), hall IO38 (10k pull-up), lights IO11 (AO3400A), solar panel JST-XH, ADC IO8 (100k/33k), JP1 = panel -> 5 V.
ELRS: ESP32-C3-MINI-1 + E28-2G4M12S (SX1280) wired as ExpressLRS "Generic C3 2400" (SCK 6, MOSI 4, MISO 5, NSS 7, BUSY 3, DIO1 1, RST 2, LED 8, button 9). CRSF: C3 TX21 -> S3 IO16, S3 IO15 -> C3 RX20.
S3 IO17 -> C3 EN (reset), S3 IO18 -> C3 IO9 (bootloader). First flash of the C3: USB pads RX USB (D-, D+, GND) or WiFi. Bind: BIND button (C3 IO9).

drive board v3: ESP32-S3 + ESC output + ELRS + INA226 + buck
Kobe Vanrobaeys
Sheet: /
File: sc02_drive.kicad_sch
Title: SC-02 floor board
Size: A1
Date: 2026-10-18
Rev: 2.0
KiCad E.D.A. 10.0.3
Id: 1/1